DA09 Sort, group, mean and median

Week 3, Tables in pandas | Session 9 of 24 | Python for Data Analysis

Today's question. Which item earns the most, and what is a typical sale in a small practice shop?

Objective. By the end: you can total a column by group, sort the result, and say when the median describes a column better than the mean.

> **WHERE THIS IS USED**
> Retail chains compare their branches by grouping sales and finding the best sellers. A typical sale and an unusually large one tell different stories. Which branch would you visit first, and why?

Before this session: DA08 Select, filter and calculate a revenue column. If you cannot explain the last example, use its error card with the instructor first.

Words for today

Word	What it means	Think of it like this
group	Gather rows with the same label before you calculate.	Sorting the sales into piles by item.
sort	Put rows in order from smallest to largest or the other way.	Lining up pupils by height.
mean	The total shared out equally: add the values and divide by how many.	Sharing a bag of mandazi equally.
median	The middle value once the values are in order.	The pupil standing in the middle of a line sorted by height.
proportion	A part divided by the whole. Times 100 it becomes a percentage.	12 pupils out of 20 passed.
denominator	The whole that a proportion is measured against.	The 20 in 12 out of 20.

Block A. Group, total and sort

groupby gathers the rows with the same item. A calculation such as sum then works inside each group. sort_values puts the answer in order.

Predict. Predict which item sold the most units and which earned the most money. Write your guess on paper first.

Your guess, written before you run it

```
import pandas as pd
shop = pd.read_csv("camp_shop.csv")
shop["revenue_ugx"] = shop["units"] * shop["unit_price_ugx"]
units_by_item = shop.groupby("item")["units"].sum()
revenue_by_item = shop.groupby("item")["revenue_ugx"].sum().sort_values(ascending=False)
print(units_by_item)
print(revenue_by_item)
```

Run it, then compare. Pen sold the most units, 103. Book earned the most money, 297000 shillings, then Juice 150000 and Pen 103000. The three revenues add to 550000. The item that sells the most units is not the item that earns the most.

Investigate.

1. Why does Pen sell the most units and earn the least money?
2. How do you check the grouped totals?

Modify. Group the revenue by date and find the date with the most revenue.

Block B. Mean, median and a proportion

Now meet the bigger Sunrise Shops table: 1513 sales from three shops. The mean can be pulled up by a few huge sales. The median is not. A proportion needs its whole named: 36 large orders out of 1513 sales.

Predict. Predict whether the mean sale value is above or below the median. Write your guess on paper first.

Your guess, written before you run it

```
sunrise = pd.read_csv("sunrise_sales.csv")
sunrise["line_value"] = sunrise["units"] * sunrise["unit_price_ugx"]
mean_value = sunrise["line_value"].mean()
median_value = sunrise["line_value"].median()
bulk = (sunrise["units"] >= 10).sum()
share = bulk / len(sunrise)
print(round(mean_value), median_value)
print(bulk, len(sunrise), round(share * 100, 1))
```

Run it, then compare. The mean sale value is about 4214 shillings and the median is 2400. A few bulk purchases pull the mean up. 36 of the 1513 sales have 10 units or more, which is 2.4 percent. The 1513 is the denominator.

Investigate.

1. Which number is a better answer to "what does a typical sale bring in"?
2. Why must you say that the whole is 1513 sales?

Modify. Find the proportion of sales with 5 units or more.

Make

Add the line_value column and total it for each branch_id in the Sunrise sales. Sort the three totals from largest to smallest. Check that the three totals add up to the overall total.

Save your work as DA09. Keep the worked example and put your own attempt below it. Write two sentences: what you changed and how you checked it.

Error card

You see	It means	Fix
KeyError: 'revenue_ugx'	You grouped a column that has not been created yet in this session.	Run the cell that creates the revenue column before the cell that groups it.
The totals do not add to the overall total	Rows were lost or counted twice before grouping.	Compare len() before and after each step to find where rows changed.

Exit ticket

A grouped total answers a question about each group, and the median stays steady when a few sales are huge.

AI use follows AI_USE_POLICY.pdf. In weeks 1 to 6 no AI-generated solutions are used for exercises, labs or build tasks. Practise for 20 to 30 minutes between classes by repeating the Make task with one value changed. If you are stuck for 30 minutes, save the error and bring it to class. A file saved on your laptop has not been shared until you use the route in START_HERE.pdf.